

Midterm Report: Measuring Performance of LLM Forecasters for Quantitative Risk Modeling of Cyber Misuse

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Abstract

AI-enabled cyberattacks are an important risk pathway for advanced AI systems, but current cyber risk frameworks rely on slow, costly human expert elicitation. This project develops a methodology for evaluating automated LLM “expert” forecasters that map benchmark performance to forecasts of real-world cyberattack capability uplift. Using forecasts over MITRE ATT&CK steps, we assess agreement between forecast variants under various experimental manipulations. Preliminary results show generally strong self-consistency across repeated runs, and broad invariance to alternative elicitation formats, expert persona prompts, and most prompt components, suggesting that LLM forecasters are not highly sensitive to these choices. Conversely, forecasters anchor strongly to provided baseline probabilities, and exclusion of baselines from the prompt results in much higher forecast variance. These findings provide a practical framework for evaluating consistency and prompt sensitivity in automated LLM forecasters, while supporting ongoing work on a scalable ground-truth benchmark for forecast accuracy.

Keywords Forecasting · Cyber risk · AI safety

1 Introduction

AI enabled cyberattacks are an important risk vector for advanced AI systems, and have already resulted in direct harm.[1] SaferAI has developed 9 cybersecurity risk models that leverage AI evaluation benchmarks as key risk indicators to forecast the degree of cyberattack capabilities uplift from frontier models.[2] Historically, the mapping between benchmark performance and forecast uplift has relied on human expert elicitation, but this process is time consuming and expensive, which imposes limits on its utility for monitoring. SaferAI would like to use automated LLM experts to elicit these forecasts for input into risk models, allowing for scalable monitoring of a wider range of models on faster time scales, but a methodology for measuring and evaluating the capability of LLM experts at this task is currently lacking.

This project seeks to develop one such methodology and to leverage that methodology to evaluate and improve LLM expert elicitation performance. This consists of three primary objectives. First, develop a metric that captures desirable characteristics of the elicited forecasts and can be used to compare various experimental manipulations of forecasters. Second, use this methodology along with other statistical tools to investigate the impact of various manipulations on the forecasting approach. And third, to develop an adjacent and scalable ‘ground truth’ accuracy that can be used to evaluate forecast quality without relying on scarce human-labeled data. This report details progress on the first two of these goals, while the third remains a future objective.

2 Methods

2.1 Forecast Elicitation

The forecasts used for development in the subsequent sections are elicited using the approach described in Barret et. al 2025. Briefly, the objective of these forecasts is to estimate the probability of success on a given MITRE ATT&CK [3] step conditional on observed success on a given benchmark task (taken from Cybench or BountyBench). These estimates are elicited at the modal (50%) level, as well as 25% and 75% levels to give some sense of the confidence interval around this estimate.

The forecast model is complex and contains numerous features which could be experimentally manipulated. The most important components for the purposes of this report include:

- **Persona descriptions:** a set of custom designed prompts intended to elicit various ‘expert personas’ during the estimation procedure.
- **Baseline risk estimates:** baseline values and confidence intervals for MITRE step risk taken from expert opinions.
- **Reasoning Prompts:** a chain of custom prompts instructing the forecaster on the desired forecast flow steps, including detailed analysis of provided benchmark task and guidance on reasoning step for translating this analysis into a forecast.

2.2 Consistency Metrics

We quantified agreement between repeated elicitation using Wasserstein distance metrics,[4] with the goal of developing a practical measure for evaluating subsequent modifications to the estimator. We focused on the first-order and second-order Wasserstein distances (W1 and W2), which summarize the discrepancy between two distributions in terms of the amount of probability mass that must be shifted to transform one distribution into the other. W1 (earthmover’s distance) captures the average magnitude of this shift, while W2 places greater weight on larger discrepancies and is correspondingly more sensitive to occasional large deviations. Confidence intervals (95%) were generated by bootstrapping distances across resampled sets of elicited points.

This metric serves two primary purposes:

- **Quantifying estimator consistency:** By comparing multiple runs with identical settings, we quantify the reliability of an estimator. Low run-to-run variability is desired, as this indicates the estimator elicits similar distributions when run on with identical data
- **Quantifying impact of experimental modifications:** If modifying some component of the estimator (e.g., prompt components) has no impact on consistency compared to the baseline approach, it strongly suggests that this modification has minimal impact on the behavior of the estimator. This can be used to hone in on specific interventions that materially impact estimator performance.

The W1 and W2 metrics are primarily computed using a linear interpolation assumption, which directly compares elicited quantiles, as this removes an additional degree of uncertainty that would be introduced by fitting these points to an assumed distribution shape. In some cases, where noted, elicited points are instead fit to a beta distribution before computing the distance using numeric integration. This provides a fairer comparison for manipulations which elicit a different number of points, as the linear approach biases these comparisons towards larger distances. In addition, we also compute discrepancy of the reported modal value and interquartile range between pairs of

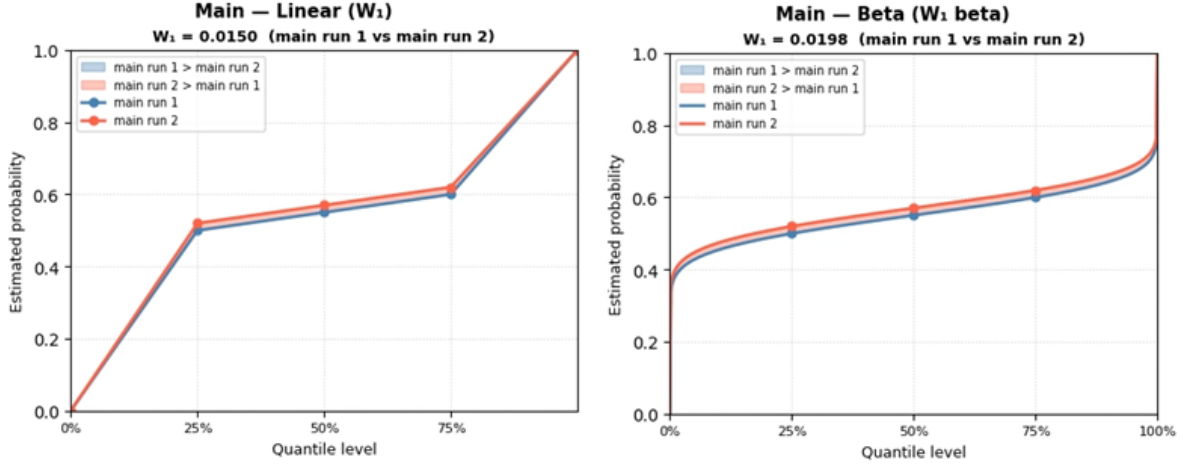


Figure 1: Example of W_1 estimation for two consecutive runs of the same estimator settings. (Left) Estimation of W_1 using a linear approach. (Right) Estimation using beta distributions fit to the same elicited points

estimates. When aggregated, these provide a separate measure of central tendency and estimate confidence across experimental variants, which are useful summary statistics for diagnosing estimator performance.

3 Preliminary Results

3.1 Baseline Consistency

We first compared the baseline self-consistency across multiple runs of the main, unmodified elicitation approach. Elicitations for these plots were conducted with a single expert persona and three MITRE steps from a single cyberattack scenario. Two models (Claude Sonnet 4.6 and GPT-5 mini) were included, and each step was run through 20 iterations resulting in 190 pairs for comparison.

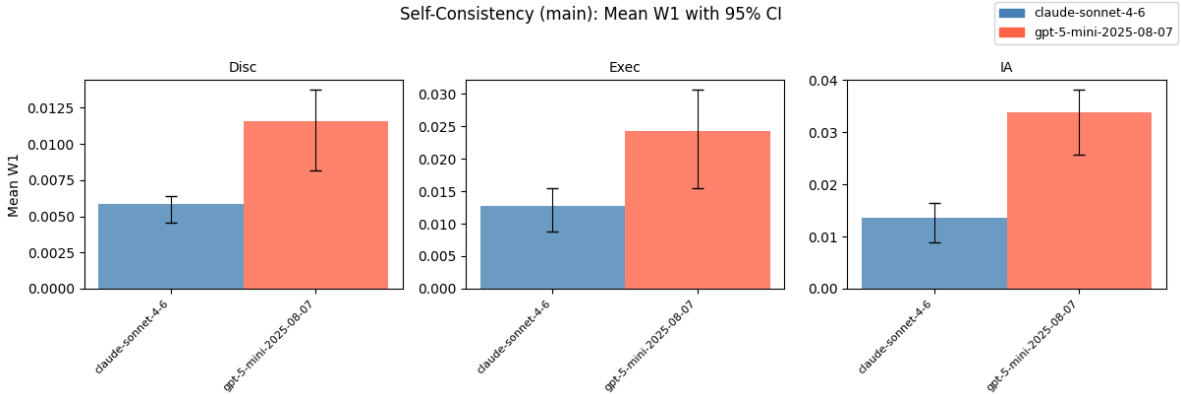


Figure 2: Mean W_1 distance and 95% confidence intervals for three MITRE steps using two estimator models.

While elicitation from GPT-5 mini had higher run-to-run average distance (indicating less agreement), both models had generally solid consistency. For reference, a W1 value of 0.05 would indicate that the probability mass was shifted an average of 5% across runs. This result is encouraging, as it suggests that there is not a large degree of stochasticity inherent to the model.

3.2 Elicitation Invariance

Aside from run-to-run consistency with identical settings, we also wanted to know whether the models remained statistically consistent when asked to elicit the same distribution in different ways. For instance, a model asked to estimate the probability of failure instead of success should report a value that is comparable to one minus the probability of success, rather than some totally different distribution.

We generated 5 variations on elicitation approach, including the negation example above and four approaches which elicited quantile values different from the 25%, 50%, and 75% points used in the main approach. All other conditions were kept identical to the ‘main’ sample runs. Elicitation points were varied both by number of points (e.g., eliciting deciles from 10-90%) and the location of points, including a pseudo-random selection of points (15%, 60%, and 77%). W1 distances were computed after fitting a beta distribution to each elicitation, as the inclusion of variable numbers of points resulted in a biased estimator when the number and location of points was mismatched.

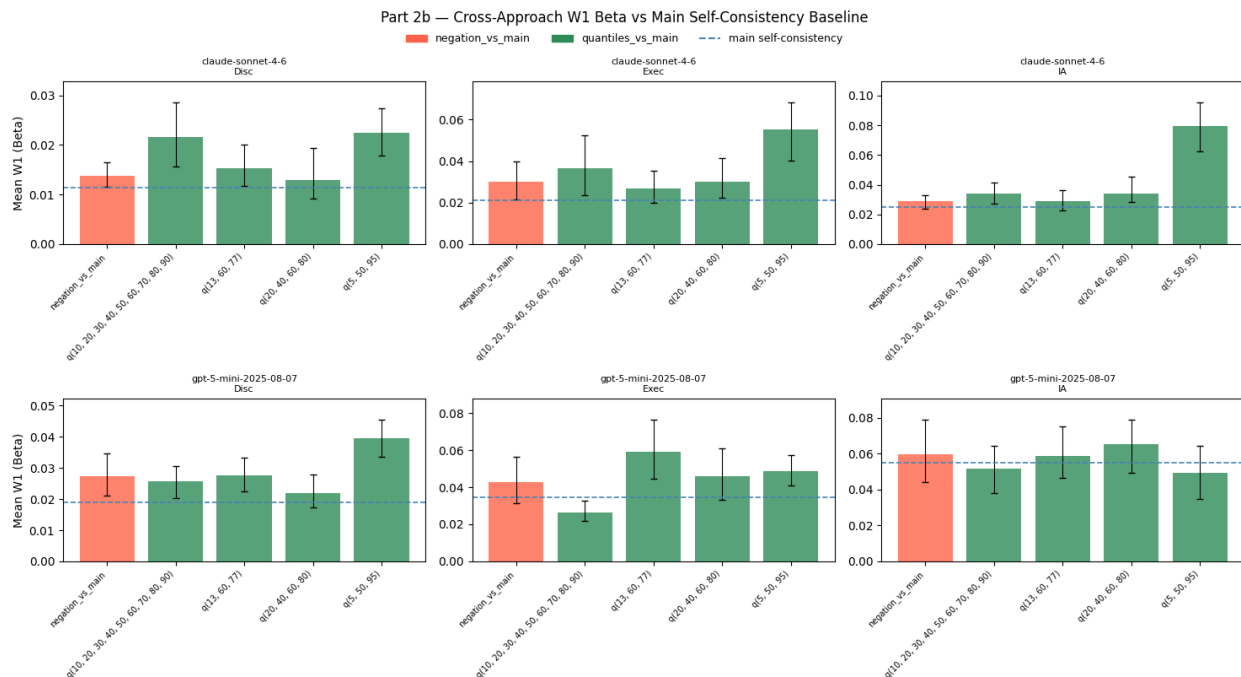


Figure 3: Mean W1 distance across approaches and 95% confidence intervals for three MITRE steps using two estimator models. W1 distance is computed between all pairs of the main approach runs and runs from various elicitation approaches. The blue reference line indicates the mean within-approach variance of the corresponding model from figure 1.

Generally, the consistency across elicitation approaches was comparable to the internal consistency in the default approach, suggesting that the model was consistent in its probability distribution estimation despite being required to report that result in diverse ways. One possible exception is

the elicitation of 5%, 50%, and 95% confidence interval points in Claude Sonnet 4.6, which appeared to meaningfully differ from the default approach across task steps.

3.3 Model choice creates more meaningful variance than expert persona

We wanted to know whether 10 expert persona we have defined create a meaningful variance and how it compares with variance created by a choice of different LLMs. We tested using Fréchet ANOVA [5] on Beta distributions fit to elicited percentiles (p25, p50, p75) via least squares (rejecting fits where $|CDF(pq)-q| > 0.05$). Three models (Claude Sonnet 4.5, GPT-4o, Gemini 2.5 Pro), 10 personas, 10 runs per persona, tested on three attack steps: TA0002 (50% baseline), TA0007 (85% baseline), T1657 (30% baseline). Within-model persona tests: Claude and GPT-4o showed no significant persona effects (ICC_F 5.7–13.9%, all $p > 0.05$). Gemini showed two significant effects at TA0002 and TA0007 (ICC_F 18.3–24.9%, $p < 0.05$) but remained smaller than cross-model effects. Cross-model tests: Model choice explained 48–69% of distributional variance (all $p < 0.001$), 3.5–12 $\times$ more than persona. Figures 4 and 5 show tight persona clustering within models versus clear model separation. This pattern carried to integer quantity estimates as well (number of actors: model ICC_F=24% vs persona ICC_F=9

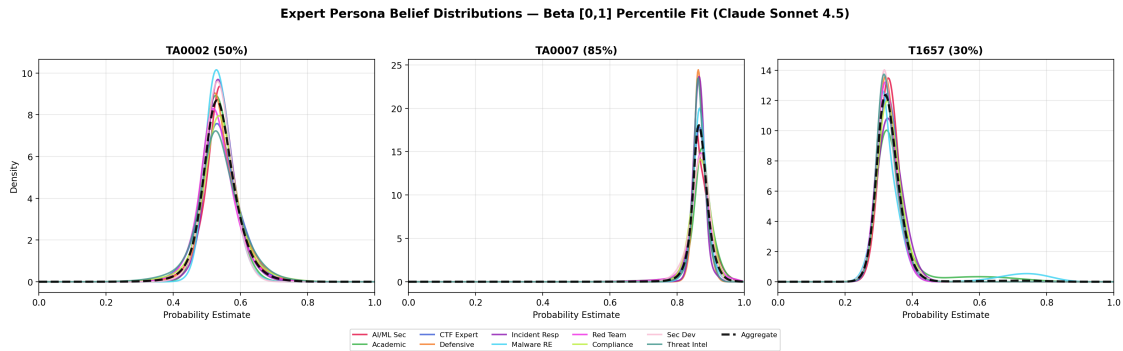


Figure 4: Persona distributions for Claude Sonnet 4.5 overlap tightly within each attack step, consistent with minimal persona-driven variance

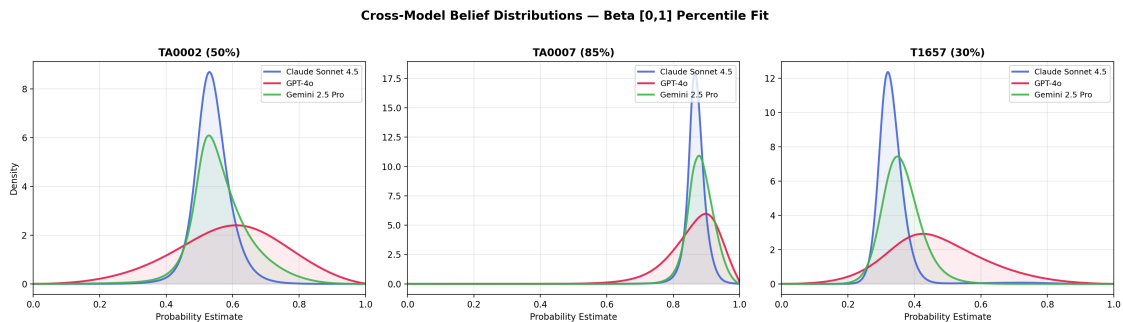


Figure 5: Cross-model distributions show clear separation across all three attack steps.

3.4 Prompt Sensitivity

We experimented with seven prompt conditions measuring W1 distance from full control prompt (approx. 615 words). We tried full prompt, no Confidence Interval (CI) from baseline, remove the

baseline, remove both CI and baseline, remove four section capability analysis, remove 3-phase reasoning scaffold, and removing all analysis + reasoning (reduced to 100 words). Preliminary experiments suggests that removing the baseline estimate (currently estimated by humans) caused the largest shift ($W1 = 0.154$, 18.1x variance compared to control). Surprisingly, trimming the entire reasoning and analysis had a lesser distribution shift and more consistent responses ($W1 = 0.016$, 0.07x variance with control).

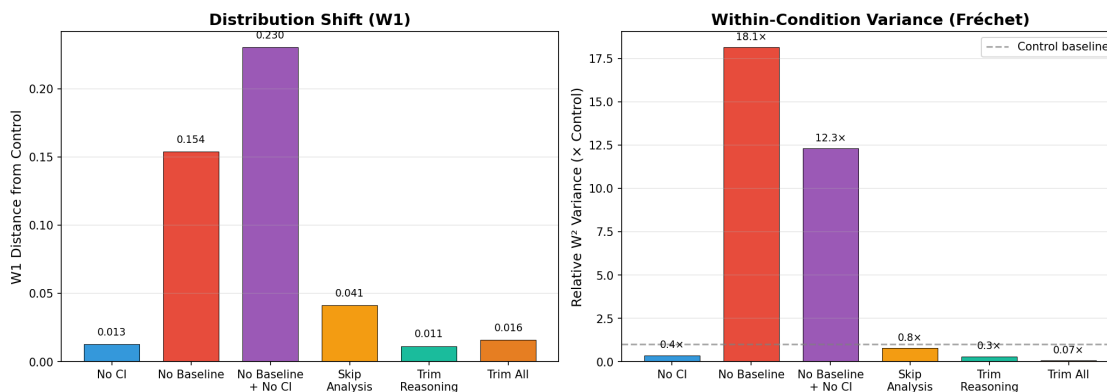


Figure 6: Impact of various prompt ablations is shown in terms of shift relative to the main approach (left) and in terms of within condition variance (right). Reference line on right plot shows the mean within condition distance of the main approach.

3.5 Baseline Anchoring

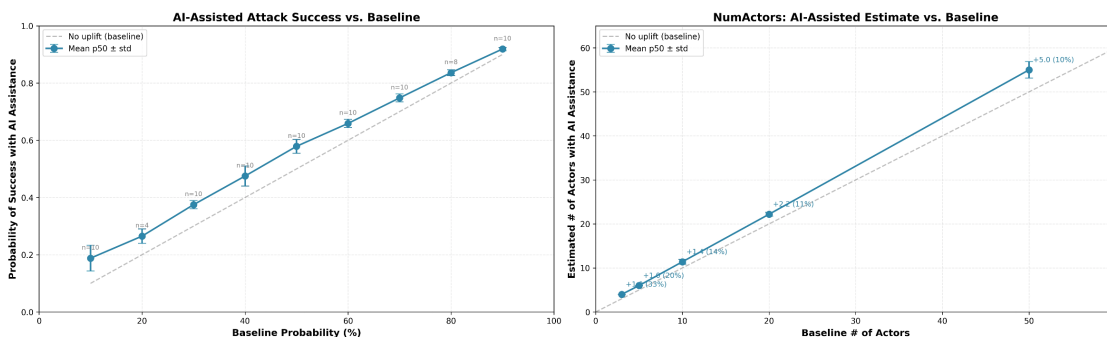


Figure 7: Results from various baseline value inputs with otherwise identical prompt settings. (Left) Across most baseline values, model appears to report a fixed level of uplift. This is attenuated when the baseline value approaches 100%. (Right) When estimating number of actors, model appears to apply a consistent proportional increase above baseline value.

To test anchoring effects, we swept baseline probability from 10% to 90% (10% increments, 10 runs per baseline) using Claude Sonnet 4.6, single persona (AI/ML Security Researcher), TA0002 (Execution step), Imaginary benchmark. Confidence intervals and baseline value removed to isolate baseline effect. The median elicited probability followed somewhat linear trend with the provided baseline (reduced uplift near higher probabilities due to ceiling effect). The median quantity node. For scenario-level quantity estimates (number of actors, unbounded), the linear relationship persisted without ceiling effects (Figure 7). This pattern followed for other steps of the attack (repeated only

on one more: Initial Access step). This suggests a strong baseline anchoring for both bounded and unbounded estimates.

4 Challenges & Open Questions

The work described here primarily addresses the consistency of elicitation and the impact of various experimental manipulations of the elicitation approach. We addressed several potential experimental interventions, but a few important open questions remain:

- Does the model respond appropriately to calibrated information on actor capability? We’ve shown that the model anchors strongly to the provided baseline value, and applies a relatively consistent uplift to this value. But it is not yet known whether differential evidence (e.g., higher or lower degrees of actor capability) impact this uplift.
- Most of these experiments were conducted in a relatively small sample space, focusing on the same three attack steps from one scenario. Future work should explore these relationships across a wider range of models, attack steps, and scenarios.

This provides a foundation for developing more accurate forecaster models, but defining this accuracy in terms of a quantifiable ground truth remains a challenge. We are addressing this challenge by developing an intra-benchmark prediction pipeline, where forecasters are asked to predict model performance on a held out set of real evaluation tasks based on their performance on a separate set of tasks. Several open questions remain for this component:

- Does model performance correlate across tasks, such that performance in one can reasonably be used to predict performance in another?
- Is it more effective to predict task level performance, or performance across a wider set of benchmark questions?
- Do the consistency principles described above also apply in the slightly different context of intra-benchmark comparison?

Finally, once this ground truth measure has been established, these principles can be used to improve performance of the LLM estimators to achieve more accurate predictions on ground truth data.

5 Next Steps

The highest priority is to establish a ground truth measure that can be used as a proxy for expected estimator performance on real cyber risk assessments. Once this measure is established, the above experiments can be repeated and extended in the new setting, and these results can be used to optimize the estimators to produce more accurate forecasts.

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